

Hall ticket No-

Code:22324

Stanley College of Engineering and Technology for Women(A)

Chapel Road , Abids

B.E I Semester (Main) Semester End Examinations March-2022

Fundamentals of Electrical Engineering

(ECE and EEE)

Time: 3 hours

Max.Marks:60

Note: Answer all questions (Compulsory)

5 X 2=10M

- 1 A Coil has a resistance of 5.2 ohm ; the resistance has to be reduced to 5 ohm by connecting a shunt across the coil. If this shunt is made of manganin wire of diameter 0.025 cm, find the length of wire required. Specific resistance for manganin is $47 \times 10^{-8} \Omega \text{ m}$. 2 M
- 2 Write the mathematical expression for a 50 Hz sinusoidal voltage of peak value 80 V. Sketch the wave form versus time t. 2 M
- 3 Mention any five applications of transformers. 2 M
- 4 Write the working principle of a capacitor start & capacitor run motor. 2 M
- 5 What do you mean by earthing of an electrical component? 2 M

Note: Answer all questions

5X10=50M

- 6 a. Three resistors are in parallel. The current in the first resistor is 0.1 A. The power dissipated in the second is 3 Watt. The voltage drop across the third is 100 V. Determine the ohmic values of resistors and the total resistance if net current is 0.2 A. 5 M
- b. State and Explain Thevenin's theorem. 5 M

OR

- c. State and Explain Superposition theorem 3M
 - d. A Wheatstone bridge ABCD is arranged as follows: $AB=10 \Omega$, $BC=30 \Omega$, $CD=15 \Omega$, $DA=20 \Omega$. A 2 V battery of internal resistance 2Ω is connected between A and C with A positive. A galvanometer of resistance 40Ω is connected between B and D. Find the magnitude and direction of galvanometer current. 7 M
- 7 a. Define the terms: (i) Real power (ii) Reactive power (iii) Apparent power and (iv) power factor. 5 M
 - b. A balanced delta -connected load of impedance $(16+j18) \Omega$ per phase is connected to 3 -phase, 440 V , 50 Hz supply. Using phasor algebra, find the line current and power absorbed by each phase. 5 M

OR

- c. Derive the relationship between line current and phase currents in case of a balanced three phase star connected sources. 5 M
- d. A pure capacitance of $318 \mu\text{C}$ is connected in series with a pure resistance of 15Ω . The circuit is supplied from a 50 Hz source and the voltage across 15Ω resistor is found to be 110 V. Calculate the supply voltage and the phase angle. 5 M

P.T.O

- 8 a. Explain the construction and working of a squirrel cage induction motor. 5 M
- b. A single phase 2KVA, 230/100V transformer has the following test results 5 M
- i) O.C. Test : 100V, 0.8A, 100W
- ii) S.C. Test : 90V, 8.7A, 300W
- Calculate
- a) Efficiency at Full load with 0.8P.F. lagging
- b) Efficiency at $\frac{1}{2}$ load with 0.8P.F. lagging
- OR**
- c. Write short notes on BH characteristics of a magnetic material. 4 M
- d. A 200 kVA, 6600/400 V, single phase transformer has 80 turns on the secondary. Calculate (i) the approximate values of primary and secondary currents (ii) the approximate number of primary turns and (iii) the maximum value of flux. Derive the necessary equation. 6 M
- 9 a. Distinguish between statically induced emf and dynamically induced emf. 3 M
- b. With a neat diagram, Explain the construction and working principle of a DC motor 7 M
- OR**
- c. Mention any three applications of (i) DC motors and (ii) single phase induction motors. 3 M
- d. With a neat diagram, Explain the construction and working principle of a DC Generator 7 M
- 10 a. Mention all the components of low tension (LT) switchgear. 5 M
- b. With help of an example explain procedure involved in calculating the energy consumed by an electrical circuit. 5 M
- OR**
- c. Write short notes on (i) Fuse (ii) MCCB and (iii) MCB. 5 M
- d. Define the term power factor, Write the typical value of power factor in practice. Is it necessary to improve the power factor? If so, Justify your answer. 5 M

Hall Ticket No-

Code: 22304

Stanley College of Engineering and Technology for Women(A)

Chapel Road , Abids

B.E I Semester (Main) Semester End Examinations March-2022

Programming for Problem Solving

(Common to all Branches)

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- 1 What is a flowchart? How it is different from an algorithm? 2 M
- 2 Write the syntax of switch statement? 2 M
- 3 Why is it necessary to give the size of an array in an array declaration? 2 M
- 4 Write the advantages and disadvantages of using pointers. 2 M
- 5 Compare and contrast text file with binary file. 2 M

PART-B

Note: Answer all questions

5 X 10= 50M

- 6 a. What are the steps involved in creating, compiling, and executing a C program? 5 M
- b. Define flow chart. List the symbols used in flow charts. 5 M

OR

- c. Define data type? Explain about different data types with examples. 5 M
 - d. Explain about bitwise operators. 5 M
- 7 a. Differentiate between if statement and if-else statement with suitable examples and proper syntax. 5 M
 - b. Write a program to find the factorial of a given integer number using 'while' loop. 5 M

OR

- c. Write a C program to check whether the given number is prime or not. 5 M
 - d. Show how break and continue statements are used in a C program, with example 5 M
- 8 a. Demonstrate different string handling functions available in C language. 5 M
 - b. Write a C program to find the largest element in an array 5 M

OR

- c. How passing parameters to function works? Explain 5 M
- d. List and explain the different types of storage class. 5 M

P.T.O

- 9 a. Explain defining and initializing a structure with an example 5 M
- b. What is pointer? How to initialize and declare pointer variables? Explain with examples. 5 M

OR

- c. How are the members of a 'Union' are initialized and accessed? 5 M
- d. List and explain the functions used to allocate and free memory dynamically. 5 M
- 10 a. Illustrate the different file opening modes that can be used with fopen(). 5 M
- b. What are the functions used for accessing files randomly? Explain with examples. 5 M

OR

- c. Devise an algorithm for linear search and explain with an illustration. 5 M
- d. Explain selection sort operation with an example 5 M

Stanley College of Engineering and Technology for Women(A)

Chapel Road , Abids

B.E I Semester (Main) Semester End Examinations March-2022

Mathematics-I

(Common to all Branches)

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2 = 10 M

- 1 Determine the nature of the series $\sum_{n=1}^{\infty} \frac{3n+1}{2n+7}$ 2 M
- 2 Find the equation of envelope of the family of curves $py + p^2x - 10 = 0$ where p is a parameter 2 M
- 3 If $x^y = y^x$ then find $\frac{dy}{dx}$, 2 M
- 4 Find the maximal directional derivative of x^3y^2z at P(1,-2,3) 2 M
- 5 Explain Bisection method. 2 M

PART-B

Note: Answer all questions

5 X 10 = 50M

- 6 (a) Test for convergence of the series $\sum_{n=1}^{\infty} \frac{n^3}{3^n}$ 5 M
(b) Discuss the convergence of the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ 5 M
OR
(c). Test for convergence of the series $\sum_{n=1}^{\infty} \frac{n!}{(n^n)^2}$ 5 M
(d). Discuss the absolute convergence of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{(n+1)}$ 5 M
- 7 (a). State and prove Rolle's mean value theorem. 5 M
(b) Discuss the applicability of Rolle's theorem for the function $f(x) = (x-4)^3(x-5)^2$ on [4,5] 5 M

OR

- (c). Find the radius of curvature of the curve $y = x^3(x-4)$ at P(4,0) 5 M
- (d). Find the Evolute of the curve $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 5 M

8 (a). If $u = F(x - y, y - z, z - x)$, then evaluate $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$ 5 M

(b). If $u = x + y + z, uv = y + z, uvw = z$ then $\frac{\partial(x, y, z)}{\partial(u, v, w)} = u^2 v$ 5 M

OR

(c). Find the maximum and minimum values of $x^3 + y^3 - 3axy$ 5 M

(d). Find The extreme value of xyz when $x + y + z = 18$ 5 M

9 (a). Show that $\nabla^2 r^n = n(n+1)r^{n-2}$. 5 M

(b). Show that $\nabla \times (\nabla f) = 0$, where f is a scalar point function. 5 M

OR

(c)&(d). Verify Greens theorem for $\oint_C (x + y) dx + x^2 dy$ where C is the triangle with vertices $(0,0), (2,0)$ and $(2,4)$ taken in that order 10 M

10 (a). Using Newton Raphson method find a root of the equation $x^3 - 2x - 5 = 0$ 5 M

(b). Using Gauss-Seidel method solve the system of equations $28x + 4y - z = 32, x + 3y + 10z = 24, 2x + 17y + 4z = 35$ 5 M

OR

(c)&(d). Find $f(x)$ from the following data using Newtons Divided difference interpolation 10M

x	-1	0	3	6	7
f(x)	3	-6	39	822	1611